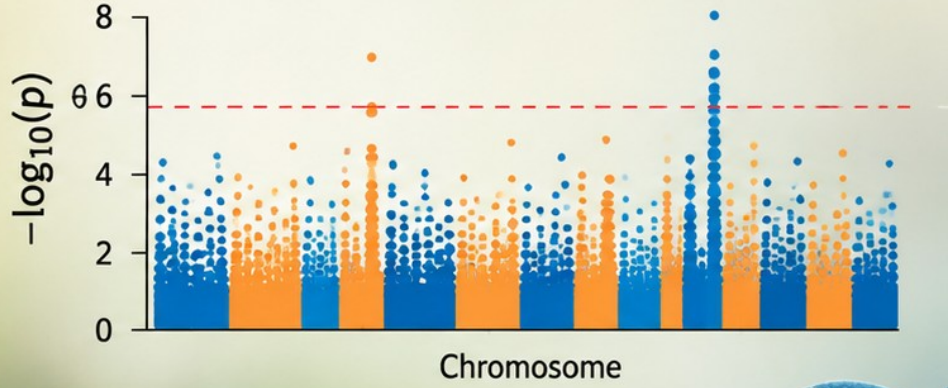
Genome-Wide Association Studies (GWAS) for Evolutionary Biology

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RAB3GAP2 is a regulator of skeletal muscle endothelial cell proliferation and associated with capillary-to-fiber ratio

Kristoffer Ström [1,2,43,45] · Nikolay Oskolkov [1,43] · Tugce Karaderi [3,4,44] · … · Cecilia M. Lindgren [3] · Paul W. Franks [5,42] · Ola Hansson [1,29] … Show more

Affiliations & Notes Article Info

Highlights

- rs115660502 in *RAB3GAP2* associates with increased skeletal muscle capillarization
- RAB3GAP2 is predominantly expressed in endothelial cells and capillaries
- Reduced RAB3GAP2 enhances endothelial proliferation and angiogenic remodeling
- Low RAB3GAP2 expression promotes TNC release and exercise-like adaptations

Summary

Skeletal muscle capillary density is correlated with physical performance and whole-body metabolic properties. Thus, we performed a genome-wide association study of skeletal muscle capillary-to-fiber ratio (C:F) ($n = 603$ males) and found that the rs115660502 G allele was associated ($p < 5 \times 10^{-8}$) with increased C:F and reduced skeletal muscle expression of RAB3 GTPase-activating non-catalytic protein subunit 2 (*RAB3GAP2*). The capillary-increasing G allele was more prevalent in elite endurance athletes than in power athletes and non-athlete controls in two independent cohorts. Low-muscle-expressing RAB3GAP2 expression quantitative trait locus (eQTL) alleles were associated with muscle damage in athletes. In healthy individuals, *RAB3GAP2* expression was reduced by high-intensity intermittent training. RAB3GAP2 protein was not uniformly expressed in muscle but predominantly expressed in the endothelium and capillaries. *RAB3GAP2* expression was lower in endurance compared with power athletes and was negatively associated with type I (oxidative) muscle fiber density. Experimental reduction of RAB3GAP2 in human endothelial cells led to (1) increased proliferation and tube formation *in vitro*, (2) regulation of secreted factors (e.g., CD70 and TNC) promoting angiogenesis and T cell activation, and (3) increased *in vivo* endothelial cell density in mice. *RAB3GAP2* expression in skeletal muscle was negatively correlated with exercise-induced release of TNC *in vivo* in humans. In conclusion, RAB3GAP2 is expressed in the microvascular endothelium and is suggested to be a negative regulator of angiogenesis through a decrease in endothelial cell proliferation, possibly mediated by RAB18, with its low-expressing variant associated with higher muscle C:F and elite endurance performance.

Graphical abstract

Figures (4) Figure Viewer

Article metrics

3 Citations 6 Captures 6 Mentions

PLUMX View details

Supplemental information (3)

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GWAS OF HISTOLOGICAL PHENOTYPES PROVIDES INSIGHTS INTO THE GENETIC ARCHITECTURE OF HUMAN SKELETAL MUSCLE

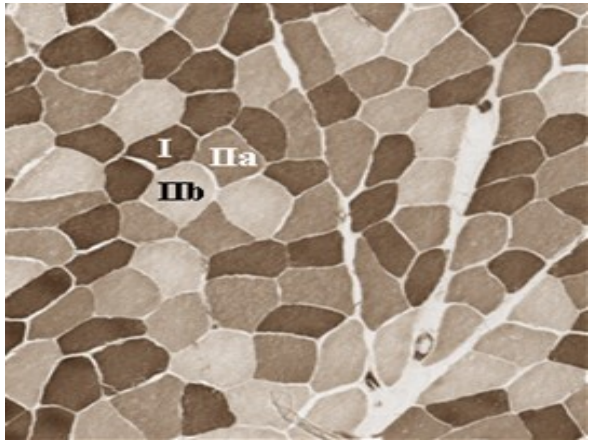
Nikolay Oskolkov



A few genetic advices on how to become Usain Bolt

LUND UNIVERSITY
DIABETES CENTRE

SKELETAL MUSCLE FIBRE TYPES

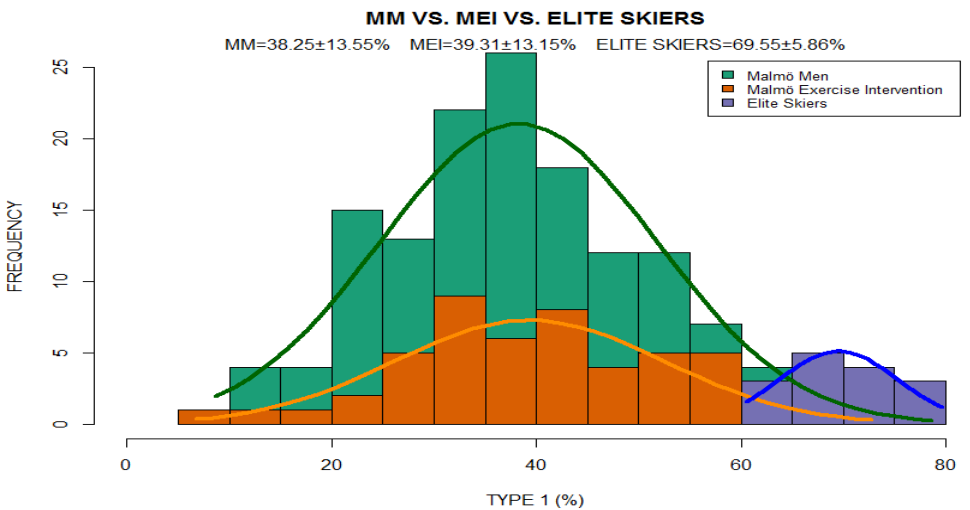


Segerström et al, J. Str. Cond. Res., 2010

	TYPE I Fibers (Red)	TYPE IIA Fibers (Red)	TYPE IIX Fibers (White)
Contraction time	Slow	Fast	Fast
Metabolism	Oxidative	Oxidative	Glycolytic
Fibre size	Small	Medium	Large

Sprinter: TYPE IIA / IIX



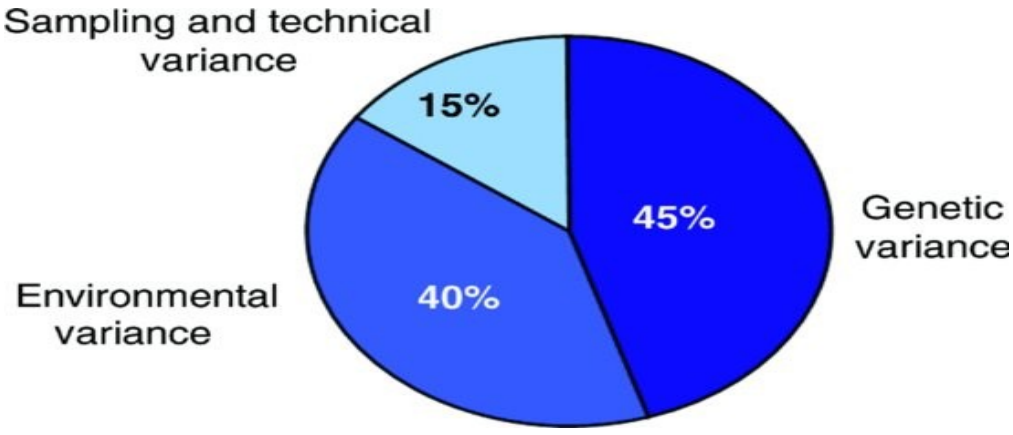


Long distance runner: TYPE I



THE ROLE OF "TALENT" IN SPORTS





J.A. Simoneau, C. Bouchard, FASEB, 1995

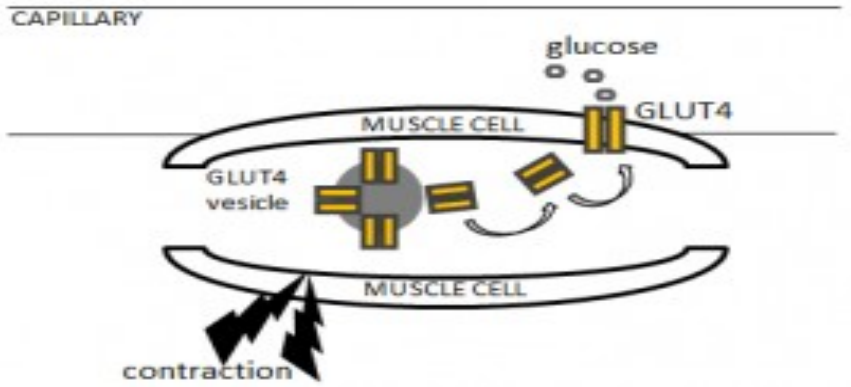




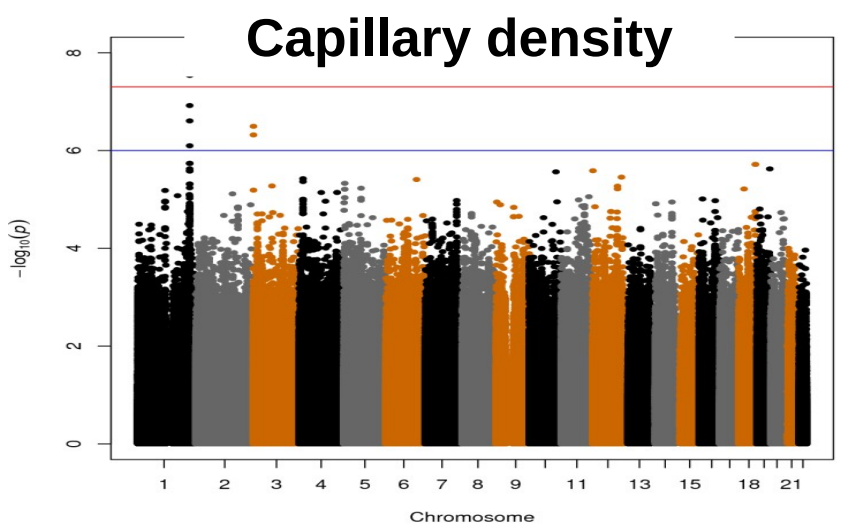
If a donkey exersices a lot it will still be a very fast and trained donkey but not a racehorse

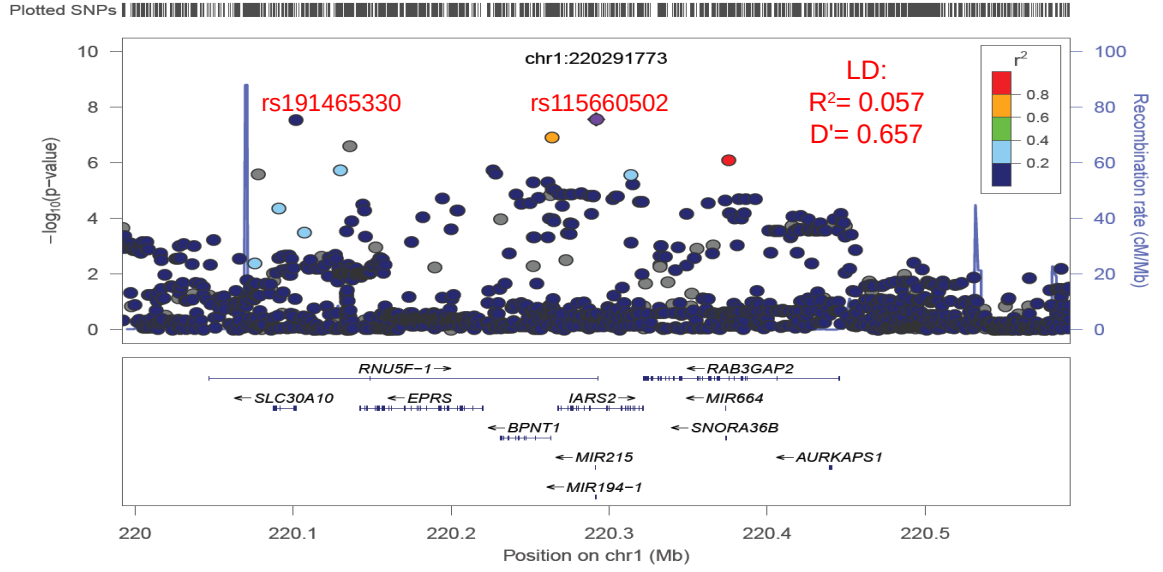
AIM:

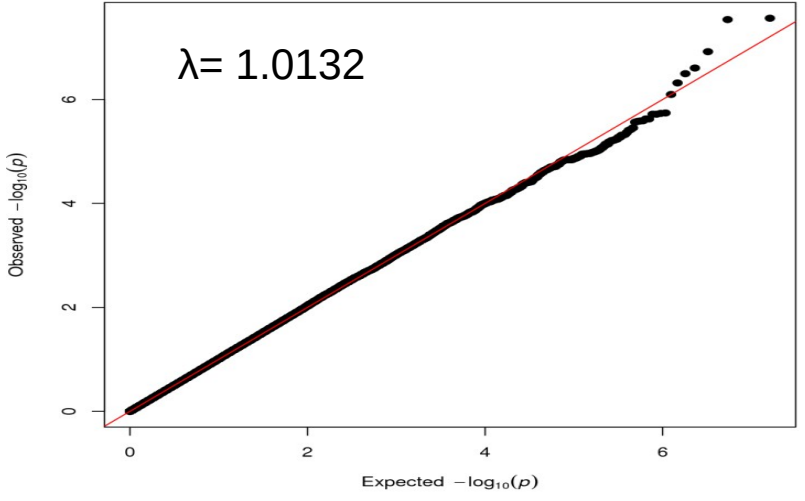
To identify genetic variation associated with skeletal muscle histological phenotypes - *e.g.* fiber types, capillary density



CAPILLARY DENSITY





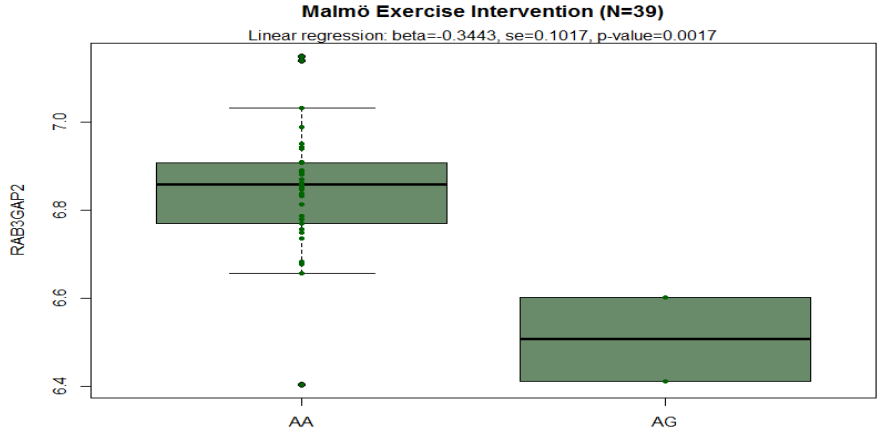


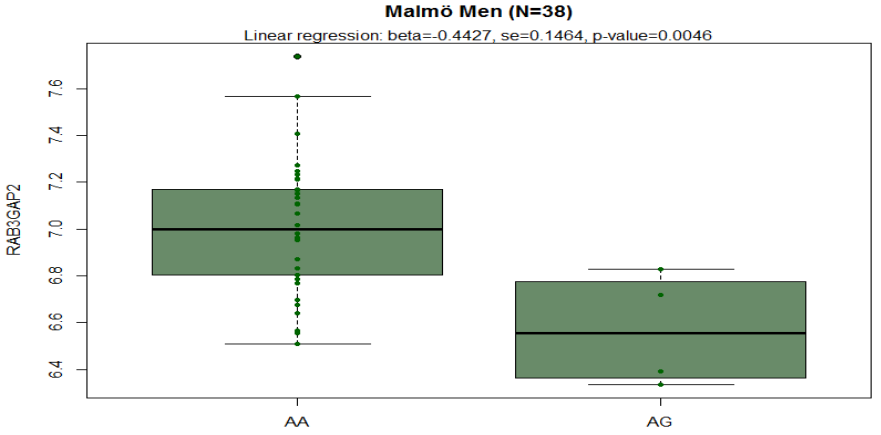
rs115660502, A/G polymorphism
MAF(ULSAM+MM+MEI)=4.8%

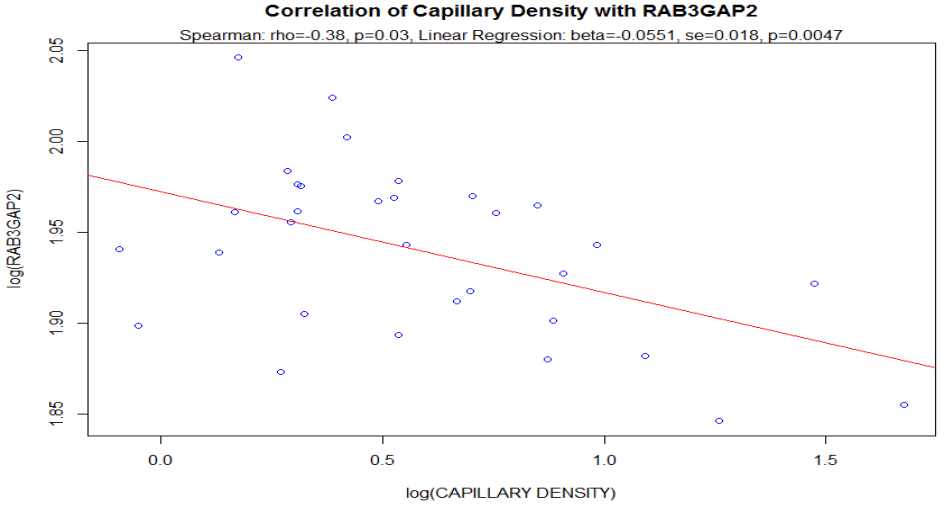
G-allele is associated with
increase of capillary density

rs115660502 REGULATES CAPILLARY DENSITY VIA RAB3GAP2

Cis-eQTL meta-analysis on human skeletal muscles: MEI+MM (77 individuals)

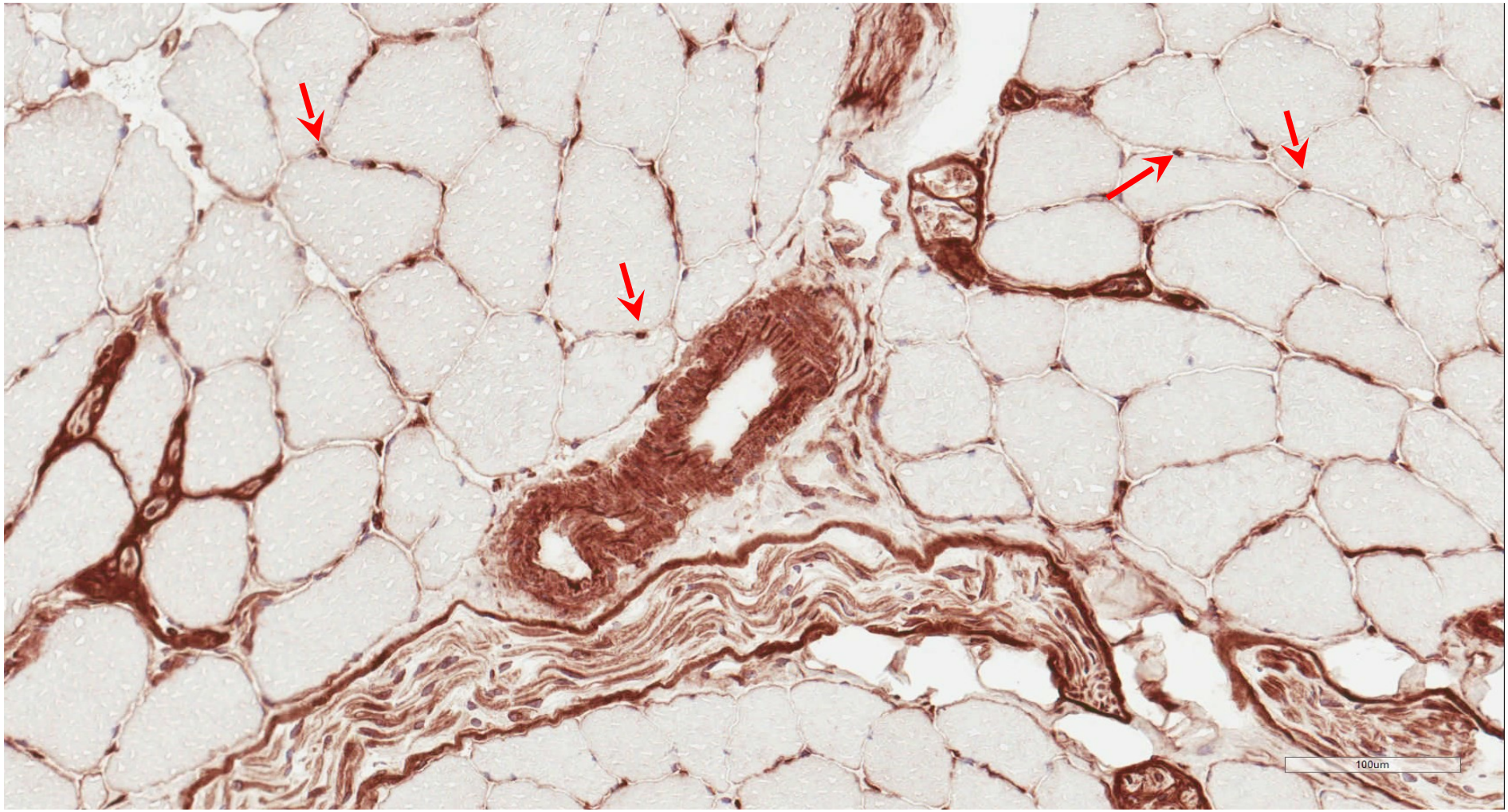






- G-allele of rs115660502 increases capillary density
- G-allele of rs115660502 decreases RAB3GAP2 gene expression in skeletal muscles
- RAB3GAP2 gene expression in skeletal muscles and capillary density are negatively correlated

IHC STAINIG FOR RAB3GAP2 IN SKELETAL MUSCLE



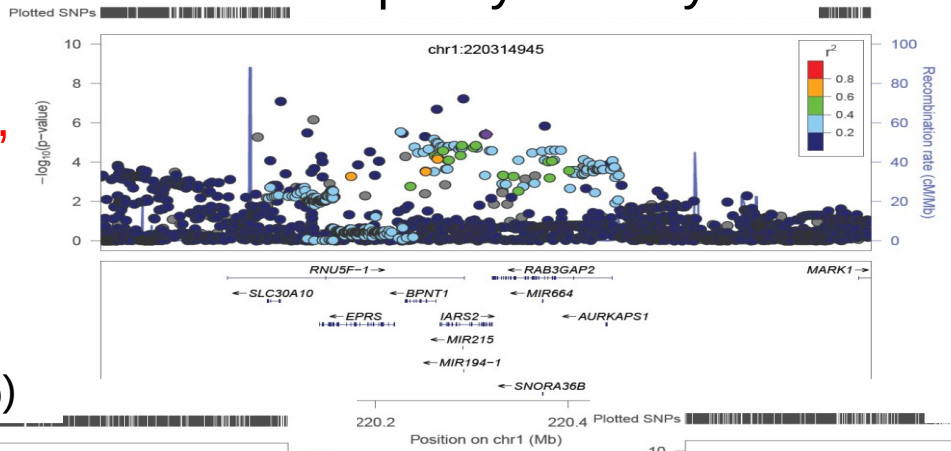
RAB3GAP2 is predominantly present in capillaries of skeletal muscles

POPULATION DIFFERENTIATION SCAN FOR POSITIVE SELECTION: CAPILLARY DENSITY AND TYPE1 (%)

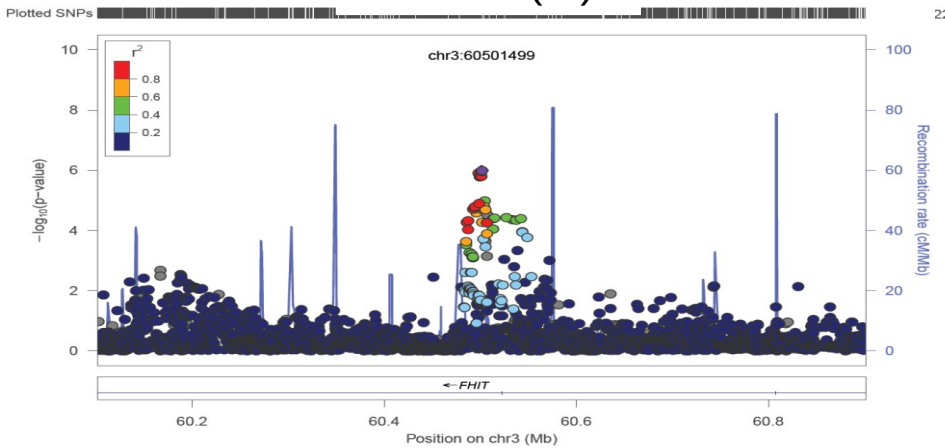
Loci-candidates for positive slection were selected by ranking by Fst and ΔDAF between EUR and AFR populations of 1000G

Capillary Density

DAF(EUR)=55%,
DAF(AFR)=85%

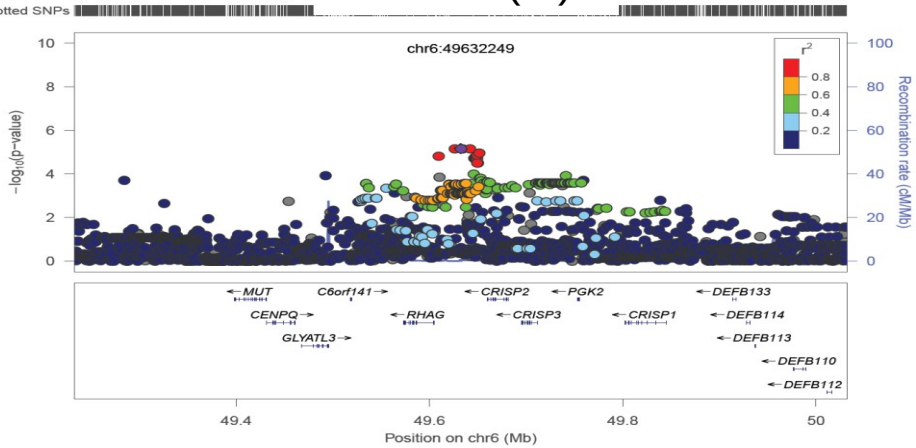


TYPE1 (%)



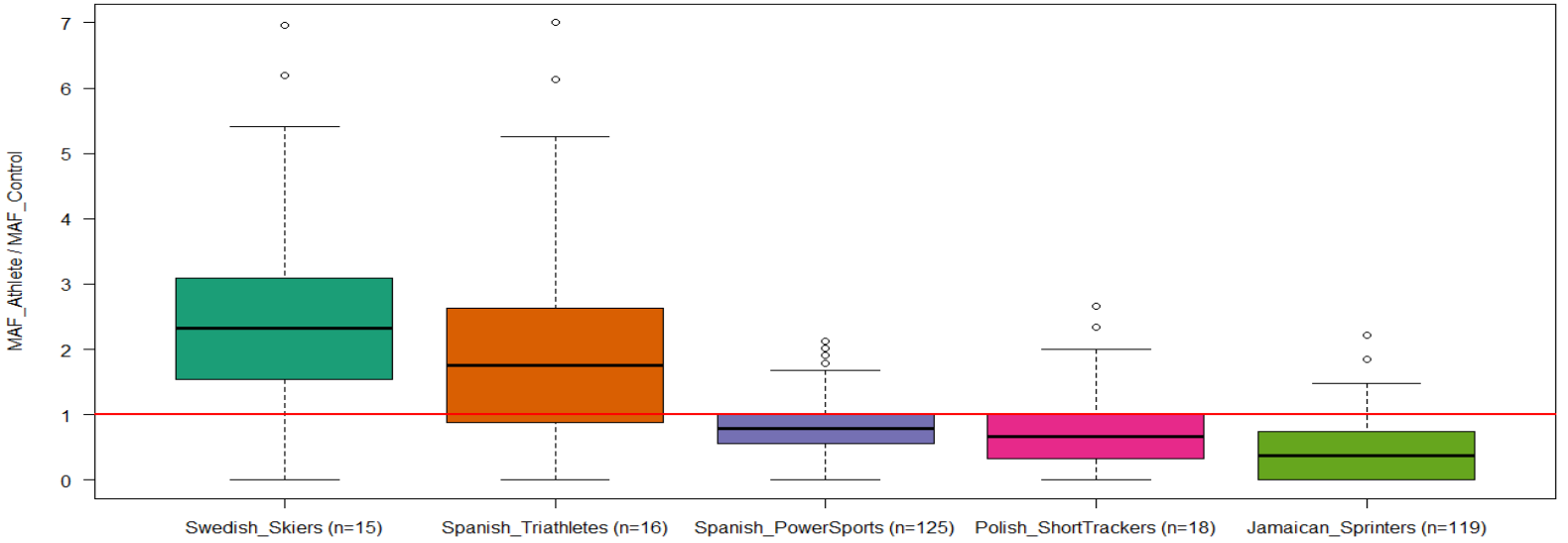
DAF(EUR)=24%,
DAF(AFR)=4%

TYPE1 (%)



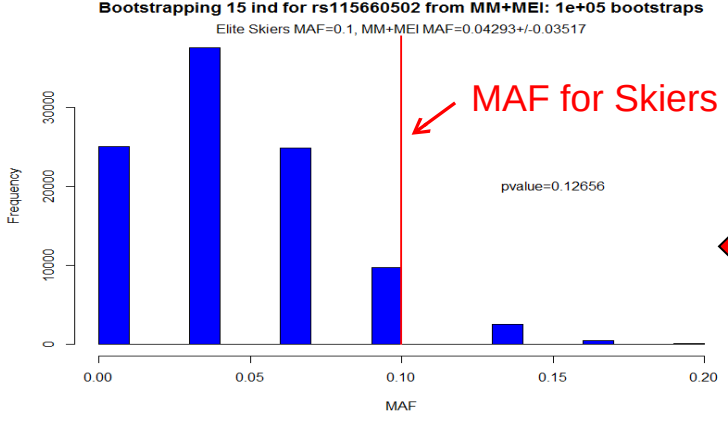
DAF(EUR)=63%,
DAF(AFR)=90%

G-ALLELE OF rs115660502 IS OVERREPRESENTED IN ENDURANCE AND UNDERREPRESENTED IN POWER SPORTS



SWEDISH CROSS-COUNTRY SKIERS: MAF=10%

SWEDISH POPULATION: MAF=4.8%



ENDURANCE POWER

MAF for Sprinters

JAMAICAN SPRINTERS: MAF=0.4%

JAMAICAN POPULATION: MAF=1.1%

